

WEEK 8 – POWER BI STUDENT PERFORMANCE DASHBOARD REPORT

1. Introduction :

Week 8 focused on creating a Student Performance Dashboard using Microsoft Power BI. The purpose of this dashboard is to analyse and present student academic performance in a simple and interactive way. The dashboard uses different visual elements such as KPI cards, charts, and slicers to represent the data clearly.

The dashboard mainly focuses on student scores in Mathematics, Reading, and Writing. It also provides comparisons based on race/ethnicity, parental level of education, lunch category, test preparation course, and gender. By using these visuals, the overall performance and differences between different student categories can be understood more easily.

2. Objectives :

The main objectives of this dashboard are:

- To analyse student academic performance using Power BI.
- To display important performance measures using KPI cards.
- To compare Mathematics scores across different race/ethnicity groups.
- To analyse student performance based on parental level of education.
- To compare student performance between Standard and Free/Reduced lunch categories.
- To compare the performance of students based on test preparation course status.
- To compare the academic performance of Female and Male students.
- To provide an interactive Race/Ethnicity slicer for filtering the dashboard.
- To present the student data in a clear and easy-to-understand visual format.

3. Dashboard Overview :

The Student Performance Dashboard contains five KPI cards, different charts, and a Race/Ethnicity slicer. The KPI cards provide a quick summary of the main performance values, while the charts provide detailed comparisons between different categories.

The dashboard allows the user to understand student performance from different perspectives without having to analyse the original dataset manually.

4. Dashboard Components :

The dashboard contains KPI cards, charts, and slicers to present the student performance data in an interactive and easy-to-understand manner.

4.1 KPI Cards :

The dashboard contains five KPI cards at the top. These cards provide a quick summary of the overall student performance.

Average Math Score:

The Average Math Score card displays **66.09**. It represents the average marks scored by all students in Mathematics. This card gives a quick understanding of the overall Mathematics performance.

Average Reading Score:

The Average Reading Score card displays **69.17**. It represents the average marks scored by students in Reading. This helps to understand the overall Reading performance.

Average Writing Score:

The Average Writing Score card displays **68.05**. It shows the overall average performance of students in Writing .

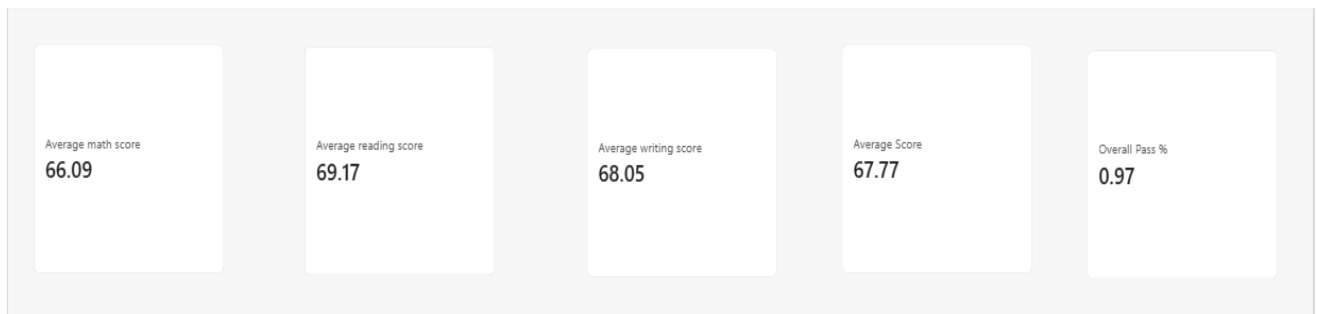
Average Score:

The Average Score card displays **67.77**. It represents the overall average score of the students and provides a general measure of their academic performance.

Overall Pass %:

The Overall Pass % card displays **0.97**. It provides a quick view of the overall passing performance represented in the dashboard.

These five KPI cards help the user understand the main performance indicators immediately without analysing every chart separately.



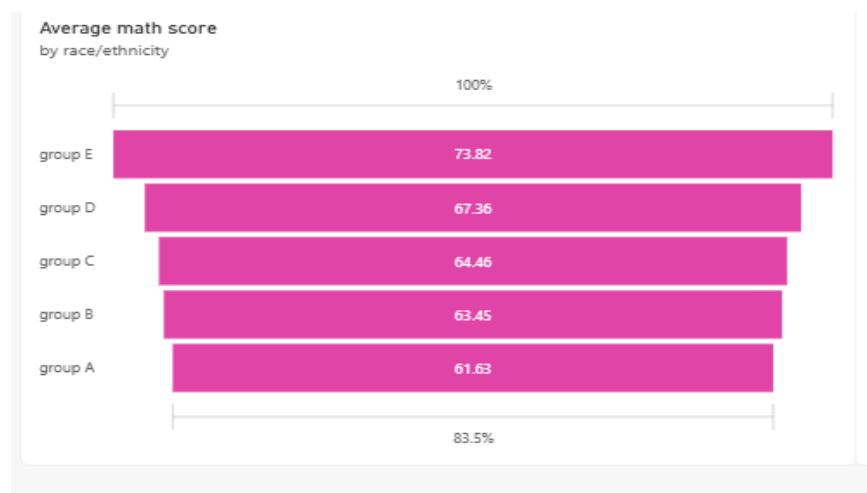
4.2 Average Math Score by Race/Ethnicity :

This visual compares the average Mathematics score across different race/ethnicity groups. The groups included are **Group A, Group B, Group C, Group D, and Group E.**

The average scores are:

- Group A – 61.63
- Group B – 63.45
- Group C – 64.46
- Group D – 67.36
- Group E – 73.82

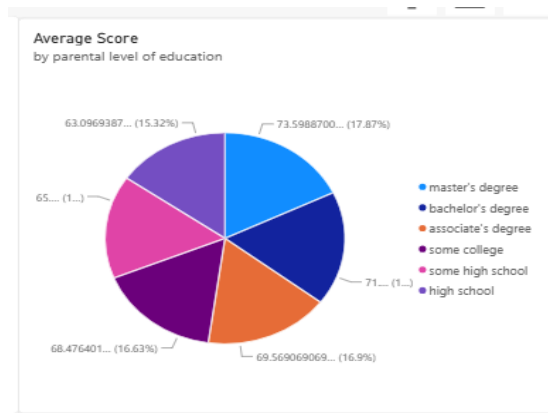
The visual helps compare Mathematics performance among the different groups. From the displayed values, Group E has the highest average Mathematics score, while Group A has the lowest.



4.3 Average Score by Parental Level of Education :

This visual represents student performance according to the **parental level of education**. The categories included are Master's degree, Bachelor's degree, Associate's degree, Some college, Some high school, and High school.

The visual shows the average score distribution for each parental education category. It helps in comparing student performance across different levels of parental education and provides a clear category-wise view of the data.



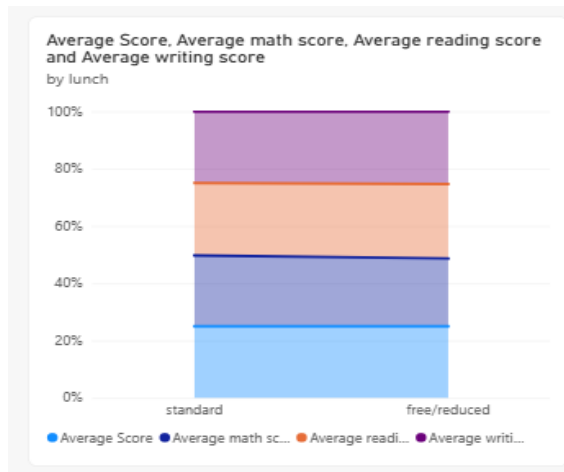
4.4 Average Score and Subject Scores by Lunch :

This visual compares student performance based on the **lunch category**. The two categories are **Standard** and **Free/Reduced**.

The visual includes:

- Average Score
- Average Math Score
- Average Reading Score
- Average Writing Score

It helps compare both overall and subject-wise performance between the two lunch categories. The user can view Mathematics, Reading, Writing, and overall scores together for easier comparison.



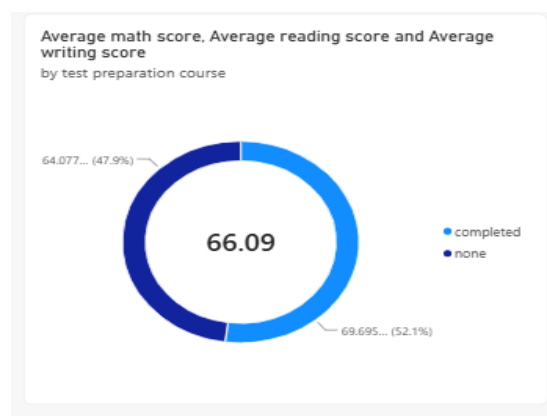
4.5 Subject Performance by Test Preparation Course :

This visual compares students based on whether they **completed the test preparation course or did not take the course.**

The categories are:

- Completed
- None

The visual uses the average Mathematics, Reading, and Writing scores to compare the performance of the two groups. It helps identify the difference in displayed subject performance between students who completed the preparation course and those who did not.



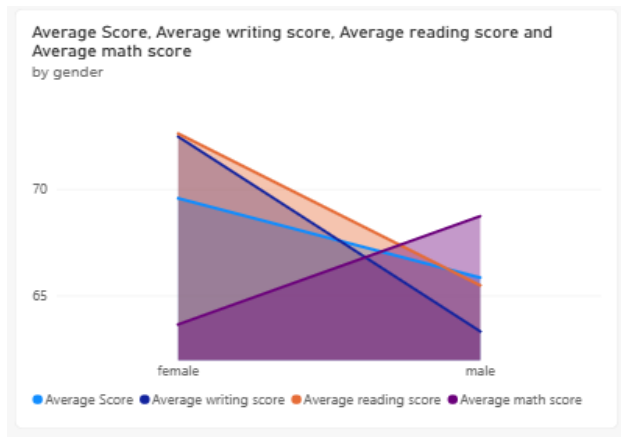
4.6 Performance by Gender :

This visual compares the academic performance of **Female** and **Male** students.

The measures included are:

- Average Score
- Average Writing Score
- Average Reading Score
- Average Math Score

It provides a simple comparison of overall and subject-wise performance between the two gender categories. This makes it easier to observe the differences in the displayed average scores.

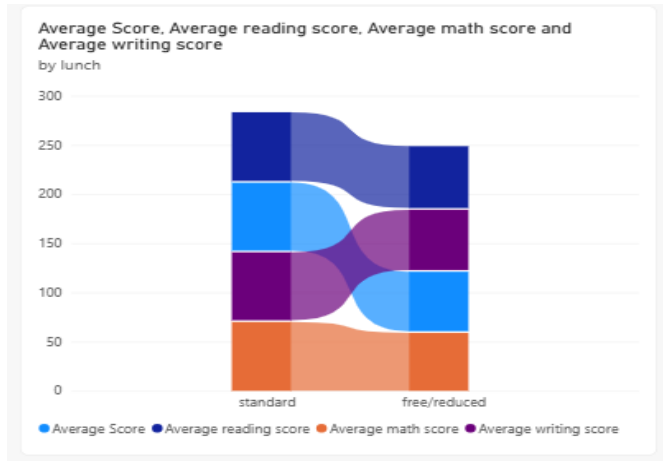


4.7 Subject Performance by Lunch :

This visual again presents subject-wise performance according to the lunch category. It compares **Standard** and **Free/Reduced** categories using:

- Average Score
- Average Reading Score
- Average Math Score
- Average Writing Score

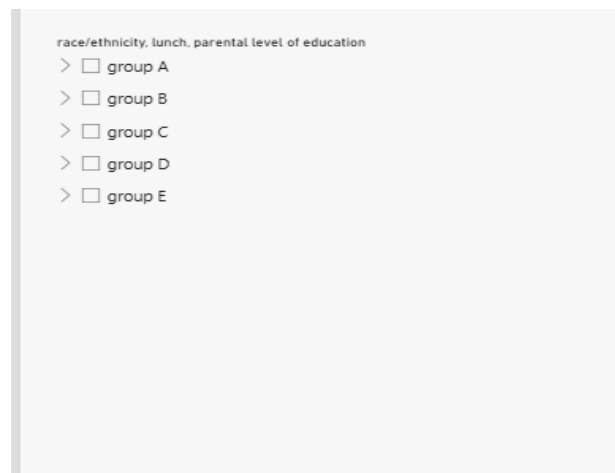
This visual makes it easier to compare the subject performance of students across the two lunch categories. Since all four measures are shown together, the performance of the two groups can be compared directly.



4.8 Race/Ethnicity Slicer :

The dashboard also contains a **Race/Ethnicity slicer** with Group A, Group B, Group C, Group D, and Group E.

The slicer allows the user to select a particular group and filter the dashboard based on that selection. This makes the dashboard interactive and allows users to focus on a specific race/ethnicity group when analysing the student performance data.



5. Key Observations :

The dashboard provides an overall view of student performance through the five KPI cards. The Average Math Score is **66.09**, the Average Reading Score is **69.17**, the Average Writing Score is **68.05**, and the Average Score is **67.77**.

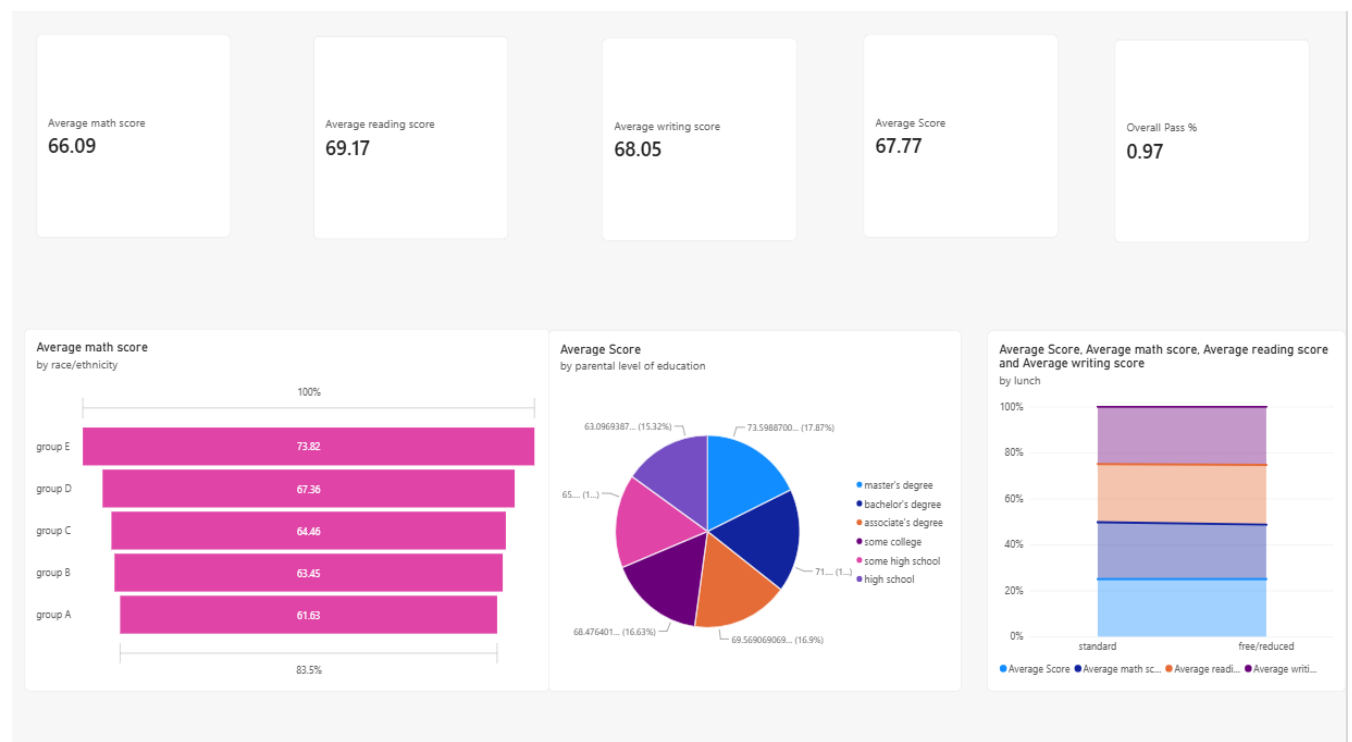
The Mathematics comparison shows differences between the five race/ethnicity groups. Group E has the highest displayed average Mathematics score of **73.82**, while Group A has the lowest displayed average of **61.63**.

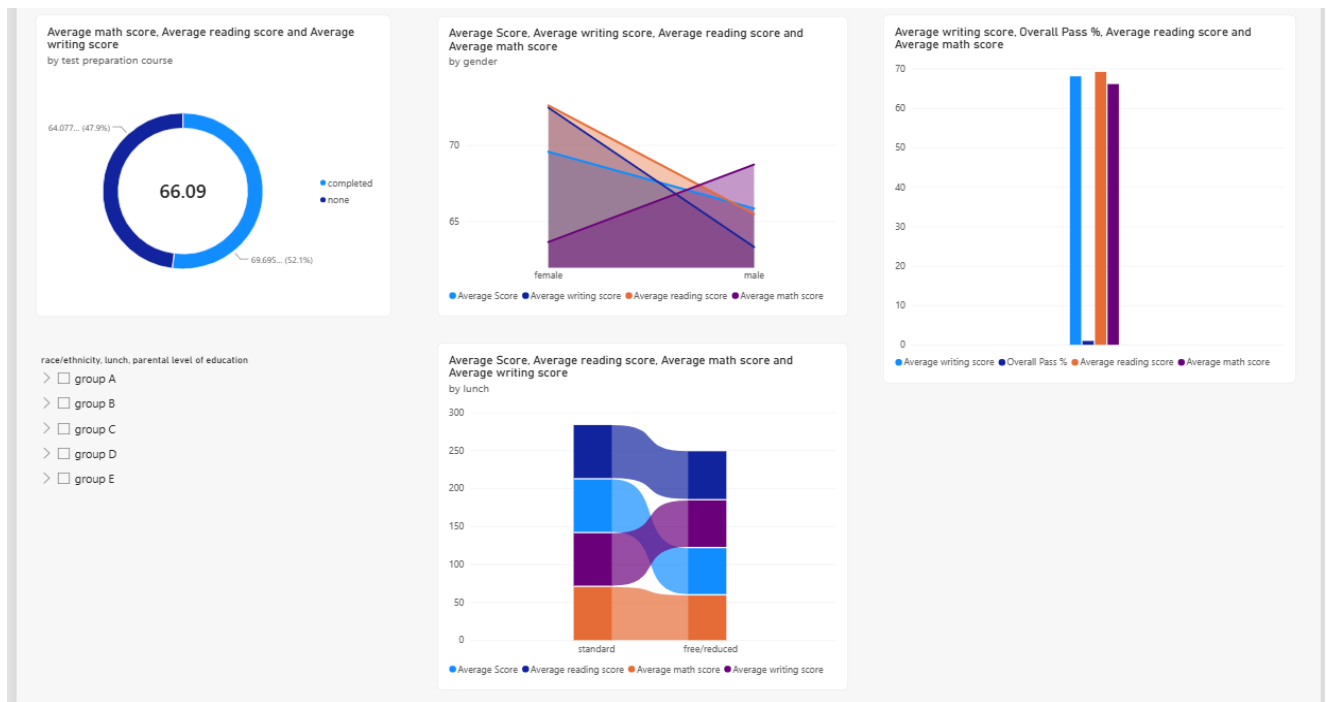
The other visuals allow student performance to be compared according to parental level of education, lunch category, test preparation course, and gender. These comparisons provide different perspectives for understanding the student performance data.

6. Learning Outcomes :

Through this Week 8 Power BI activity, I learned how to create a dashboard using different types of visual elements. I understood how KPI cards can be used to display important summary values and how charts can be used to compare different categories.

I also learned how slicers can make a dashboard interactive by allowing users to filter the data according to their requirements. This activity improved my understanding of data visualisation, comparison of datasets, and presenting analytical results in a simple and understandable format.





7. Conclusion :

The Week 8 Student Performance Dashboard successfully presents student academic performance using Power BI. The dashboard combines KPI cards, charts, and a Race/Ethnicity slicer to provide a clear and interactive view of the data.

The KPI cards give a quick summary of Mathematics, Reading, Writing, Average Score, and Overall Pass Percentage. The different charts provide comparisons based on race/ethnicity, parental level of education, lunch category, test preparation course, and gender. The Race/Ethnicity slicer further improves the dashboard by allowing users to analyse specific groups.

Overall, this dashboard demonstrates how Power BI can be used to transform student performance data into meaningful visual information. It makes the data easier to understand, compare, and analyse through an interactive dashboard.

Submitted By :

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